

Table 1*

Statistics for Selecting the Lag Order of the Earnings Equation

	With Deterministic Trends					Without Deterministic Trends			
p	AIC	SBC	$\chi^2_{SC}(1)$	$\chi^2_{SC}(4)$		AIC	SBC	$\chi^2_{SC}(1)$	$\chi^2_{SC}(4)$
1	319.33	302.14	16.86 [†]	35.89 [†]		317.51	301.64	18.38 [†]	34.88 [†]
2	324.25	301.77	2.16	19.71 [†]		323.77	302.62	1.98	21.52 [†]
3	321.51	293.74	0.52	17.07 [†]		320.87	294.43	1.56	19.35 [†]
4	334.37	301.31	3.48**	7.79**		335.37	303.63	3.41**	7.13
5	335.84	297.50	0.03	2.50		336.49	299.47	0.03	2.15
6	337.06	293.42	0.85	3.58		337.03	294.72	0.99	3.99
7	336.96	288.04	0.17	2.20		336.85	289.25	0.09	0.64

* Notes: p is the lag order of the conditional model (5.5), with zero restrictions on the coefficients of lagged changes in the productivity variable. $AIC_p = LL_p - s_p$ and $SBC_p = LL_p - \frac{s_p}{2} \ln(T)$, are the Akaike and Schwarz Information Criteria, where LL_p is the maximized log-likelihood value of the model, p is the lag order, s_p is the number of freely estimated coefficients, and T is the sample size. $\chi^2_{SC}(1)$ and $\chi^2_{SC}(4)$ are the LM statistics for testing residual serial correlations of orders 1 and 4. The symbols [†], *, and ** represent significance at 1% or less, 5% or less, and 10% or less, respectively.

Table 2*

F- and t- Statistics for Testing the Existence of a Long-Run Earnings Equation

	With Deterministic Trends			Without Deterministic Trends	
p	F_{IV}	F_V	t_V	F_{III}	t_{III}
4	2.99 ^a	2.34 ^a	-2.26 ^a	3.63 ^b	-3.02 ^b
5	4.42 ^b	3.96 ^b	-2.83 ^a	5.23 ^c	-4.00 ^c
6	4.78 ^c	3.59 ^b	-2.44 ^a	5.42 ^c	-3.48 ^b

* Notes: p is the lag order of the underlying model. See also the notes to Table 1. F_{IV} is the F-statistic for testing zero restrictions on the coefficients of the lagged level variables and the trend term in (5.5). F_V is the F-statistic for testing zero restrictions on the coefficients of the lagged level variables in (5.5). F_{III} is the F-statistic for testing zero restrictions on the coefficients of the lagged level variables in (5.5) without the trend term. t_V and t_{III} are the t-ratios of the coefficient of w_{t-1} in (5.5) with and without a deterministic linear trend. a denotes that the statistic lies below the 95% lower bound, b denotes it falls within the 95% bounds, and c denotes that it falls outside the 95% upper bound.

The various statistics in Table 2 need to be compared with the critical value bounds provided in Tables C1 and C2. First consider the bounds F test. For the model with a deterministic trend, F_V is the standard F-statistic for testing the restrictions $\pi_{ww} = 0$ and $\pi_{wx.x} = \mathbf{0}$, while F_{IV} is the standard F-statistic for testing $\pi_{ww} = 0$, $\pi_{wx.x} = \mathbf{0}$, and $c_1 = 0$, in (5.5). As has been argued in Pesaran, Shin and Smith (1998), the statistic F_{IV} which sets the trend coefficient to zero under the null of no level long-run relationship is more appropriate than F_V which does not impose such a restriction. Notice that when the trend coefficients are not restricted, (5.5) implies a quadratic trend in real wages under the null hypothesis of $\pi_{ww} = 0$ and $\pi_{wx.x} = \mathbf{0}$, which is not plausible. The critical value bounds for the statistics F_{IV} and F_V are given in Tables C1.iv and C1.v, respectively. Since the model contains 4 regressors, the 95% critical value bounds are (3.66, 4.76) and (3.47, 4.57) for F_{IV} and F_V , respectively. The test outcome critically depends on the choice of the lag order, p . For $p = 4$ the hypothesis that there exists no long-run earnings equation is not rejected at the 95% level, irrespective of whether the regressors are $I(0)$ or $I(1)$. For $p = 5$ the bounds test is inconclusive. For $p = 6$ (selected by the AIC) the statistic F_V is still inconclusive, but $F_{IV} = 4.78$ just lies outside the 95% critical value bounds and rejects the null hypothesis that there exists no long-run earnings equation, irrespective of whether the regressors are $I(0)$ or $I(1)$.¹⁷ This conclusion is confirmed even more conclusively when the bounds F-test is applied to the earnings equations without a linear trend. The relevant test statistic is F_{III} , and its associated 95% critical value bounds are (2.86, 4.01).¹⁸ For $p = 4$, $F_{III} = 3.63$, and the test result is inconclusive. But for $p = 5$ and 6 the values of F_{III} are 5.23 and 5.42 and the hypothesis of no long-run earnings equation is conclusively rejected.

The results from the application of the bounds t-tests to the earnings equations are less clear cut and do not allow the imposition of the trend restrictions discussed above. The two t-statistics reported in Table 2, t_V and t_{III} , are the t-ratios of the OLS estimate of π_{ww} in (5.5), with and without a linear time trend, respectively.¹⁹ The 95% critical value bounds for t_{III} and t_V tests are (-2.86, -3.99) and (-3.41, -4.36).²⁰ Therefore, when a linear trend is included in the model the bounds t-test does not reject the null hypothesis even for $p = 5$ or 6. But when the trend term is excluded the null hypothesis is just rejected for $p = 5$.

Overall, the test results support the existence of a long-run earnings equation when a sufficiently high lag order is selected and when the statistically insignificant deterministic trend term is excluded from the conditional model. Such a specification is in accord with the evidence on the performance of the alternative conditional models set out in Table 1, and in the remainder of this section we focus our attention on this specification and provide estimates of the long-run coefficients and the short-run dynamics based on a conditional earnings equation with $p = 6$, but without a deterministic time trend. For this model, using the ARDL approach to the estimation of the long-run relations discussed in Pesaran and Shin (1999), we obtain the following level long-run earnings equation²¹

¹⁷The same conclusion is also reached for $p = 7$.

¹⁸See Table C1.iii.

¹⁹Notice also that lagged changes in the productivity variable are excluded.

²⁰See Tables C2.iii and C2.v, for $k = 4$ under the columns headed 95%.

²¹Notice that the ARDL approach advanced in Pesaran and Shin (1999) is applicable irrespective of whether the regressors are $I(0)$ or $I(1)$.

$$w_t = \begin{matrix} 1.063 & -0.114 & -0.998 & +1.495 & +2.714 & +\hat{v}_t, \\ (0.0526) & (0.0398) & (0.0297) & (0.377) & (0.274) & \end{matrix} \quad (5.6)$$

where $\hat{v}_t$ is the error-correction term. The standard errors of the long-run estimates are given in brackets.²² All the long-run estimates are highly significant and have the expected signs. The coefficients of the productivity and the wedge variables do not differ significantly from unity. In Treasury's earnings equation the long-run coefficient of the productivity variable is imposed to be unity, and the above estimates can be viewed as providing empirical support for such an *a priori* restriction. Our long-run estimates of the effects of the unemployment rate and the union variable on real wages (namely -0.114 and 1.495) are also in line with Treasury estimates of -.09 and 1.31.²³ The main difference between the two sets of estimates concerns the long-run coefficient of the wedge variable. We obtain a much larger estimate, almost twice as much as the estimate obtained by the Treasury.

The error correction regression associated with the above (level) long-run relationship is given in Table 3.²⁴ These estimates provide further direct evidence on the complicated dynamics that seem to exist between real wage movements and its main determinants.²⁵ All the five lagged changes in real wages are statistically significant, further justifying the choice of $p = 6$. The error correction coefficient is estimated to be -0.235 (0.0675),²⁶ which is reasonably large and highly significant. The auxiliary equation of the autoregressive part of the model has the real roots 0.9258 , and -0.8931 , and two pairs of complex roots with moduli, 0.7853 , and 0.5951 ; thus suggesting an initially cyclical real wage process which slowly converges towards its equilibrium given by (5.6).²⁷ Despite the many insignificant coefficients that are retained in this error correction specification, the regression fits reasonably well and satisfies the diagnostic tests for non-normal errors and heteroskedasticity. However, it fails the functional form misspecification at the 5% level; which may be suggestive of some non-linear effects or asymmetries in the adjustment of real wage process that our linear specification is incapable of taking into account.²⁸ Recursive estimation of the

²²The long-run estimates and their standard errors are computed using *Microfit 4.0*. See Pesaran and Pesaran (1997).

²³CSW do not report standard errors for the long-run estimates of the Treasury earnings equation.

²⁴In practice it may be desirable also to derive a more parsimonious error correction model by imposing a unit long-run coefficient on the productivity variable and by dropping lagged changes with (jointly) statistically insignificant coefficients. But for our purposes this does not seem to be necessary.

²⁵The standard errors of the estimates reported in Table 3 allow for the uncertainty associated with the estimation of the long-run coefficients. This is important in the present application where it is not known with certainty whether the regressors are $I(0)$ or $I(1)$. Only in the case where it is not known for sure that all the regressors are $I(1)$ and cointegrated would it be reasonable in large samples to treat the estimates of the long-run level coefficients as known; on the grounds of their super-consistency.

²⁶The error correction coefficient in the Treasury's earnings equation is estimated to be -0.1848 (0.0528), which is quite a bit smaller than our estimate. (See p. 11 in Annex of CSW.) This seems to be due to the shorter lag lengths used in the estimation of the Treasury's equation rather than the fact that it has been estimated over a shorter time period: 1971q1-1994q3. Notice also that the t-ratio reported for this coefficient does not have the standard t-distribution.

²⁷The complex roots are $0.3406 \pm 0.7076i$, and $-0.2016 \pm 0.5859i$, where $i = \sqrt{-1}$.

²⁸The error correction regression in Table 3 also passes the residual serial correlation test. However, the